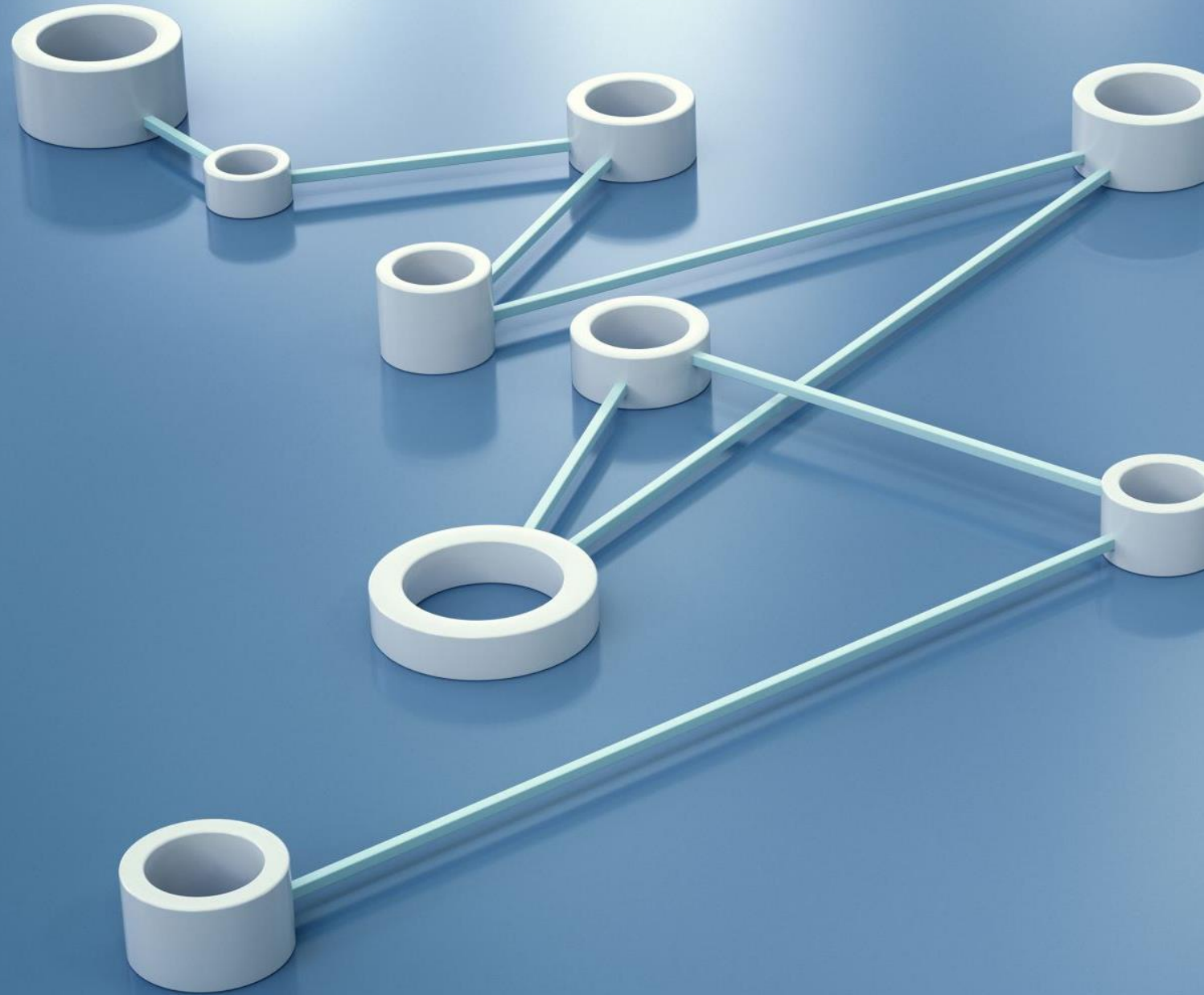

INTRODUCTION TO PROBABILISTIC DATA LINKAGE



WHY RECORD LINKAGE?

- In a perfect world, we would have a single, unique identifier, for every individual
- In this world, we often lack truly unique identifiers
 - Within datasets: for all antenatal appointments, we do not have a unique identifier for each patient
 - Between datasets: when joining antenatal appointments to deliveries, we do not have a unique identifier for each patient
- **Goal of record linkage:** generate a unique identifier
 - Can be performed within datasets (deduplication) and/or between datasets (linkage)

WHY PROBABILISTIC LINKAGE?

- **Deterministic linkage:** join results using rules

Name	Date of birth	Postcode	Phone number
Cecina Babich Morrow	29/03/1997	BS20EH	
Cecina Babich	29/03/1997		5555555555
Crcina Babich Mprrow	29/03/2097	BS20EH	5555555555

Rules

1. If name and date of birth match, then match
 2. If first name and date of birth and postcode match, then match
 3. Otherwise, no match
-

WHY PROBABILISTIC LINKAGE?

- **Probabilistic linkage:** join results using evidence
 - The evidence of a match is commonly represented as a probability

Name	Date of birth	Postcode	Phone number
Cecina Babich Morrow	29/03/1997	BS20EH	
Cecina Babich	29/03/1997		5555555555
Crcina Babich Mprrow	29/03/2097	BS20EH	5555555555

- Using this approach, we could calculate the probability that each pair of records is a match: according to fastLink, there is 100% probability each of these records is a match
 - For an interactive visualisation of this process: https://www.robinlinacre.com/intro_to_probabilistic_linkage/
-

FELLEGI-SUNTER MODEL

Main model for probabilistic linkage (1969)

PARAMETERS

- Three parameters:
 - λ : the probability that any two records match
 - If you don't know anything, how likely is a match?
 - m : the probability of a given observation given the records *are* a match
 - E.g. if two records belong to the same people, what is the probability that date of birth is the same?
 - u : the probability of a given observation given the records *are not* a match
 - E.g. if two records do not belong to the same people, what is the probability that date of birth is the same?
-

PARAMETERS

m probability

A measure of **data quality/reliability**.

How often might a person's information change legitimately or through data error?

- **Names:** typos, aliases, nicknames, middle names, married names etc.
- **DOB:** typos, estimates (e.g. 1st Jan YYYY where date not known)
- **Address:** formatting issues, moving house, multiple addresses, temporary addresses

u probability

A measure of **coincidence/cardinality**¹.

How many different people might share a given identifier?

- **DOB** (high cardinality) – for a flat age distribution spanning ~30 years, there are ~10,000 DOBs (0.01% chance of a match)
 - **Sex** (low cardinality) – only 2 potential values (~50% chance of a match)
-

MATCH WEIGHTS

- **Match weight:** score assigned to a piece of information (e.g. first name) which measures the amount of evidence of a match
 - Based on the relative size of m and u

MATCH WEIGHTS

When fields agree, we add points:

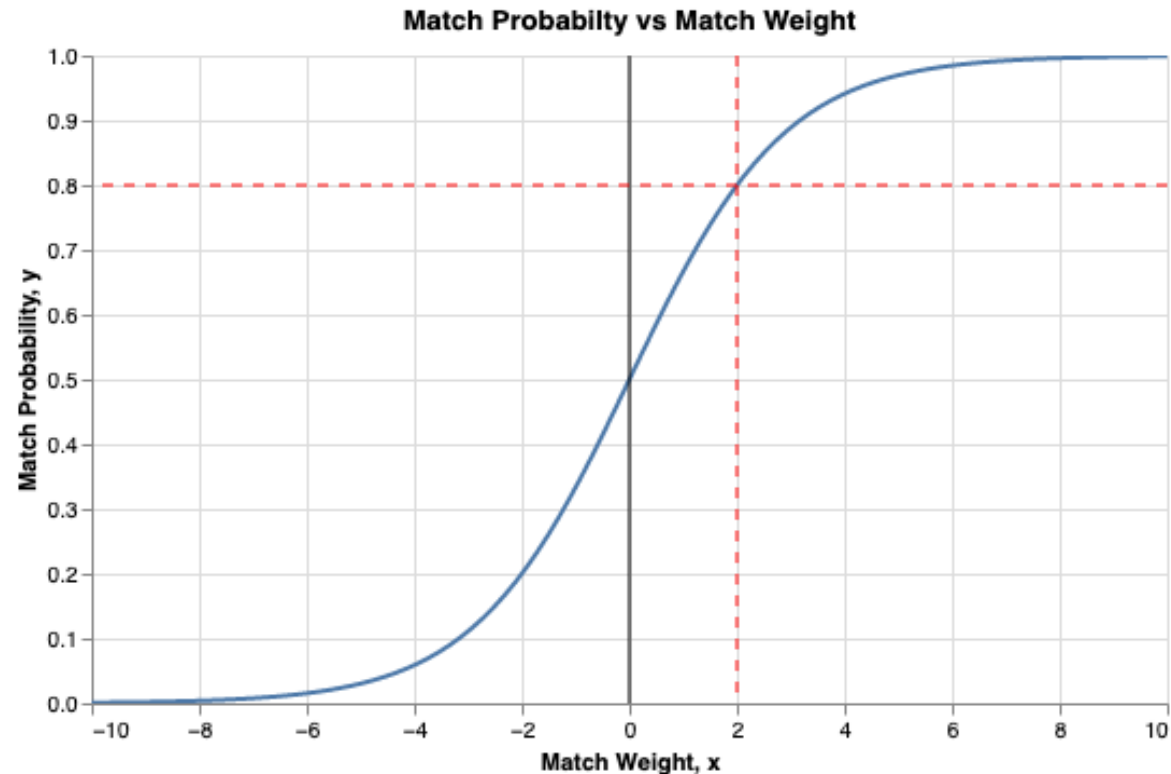
	Low u (rare data)	High u (common data)
High m (reliable data)	Large positive weight	Small positive weight
Low m (unreliable data)	Small positive weight	Negligible weight

When fields disagree, we subtract points:

	Low u (rare data)	High u (common data)
High m (reliable data)	Large negative weight	Small negative weight
Low m (unreliable data)	Small negative weight	Negligible weight

MATCH PROBABILITIES

- **Match probability:** the probability that two records are a match given the observation
 - Can be calculated from match weight



GRAPHS

[Linked data as graphs](#)

<https://github.com/babichmorrowc/bristol-uganda-data-week/tree/main/intro-to-probabilistic-linkage>

DATA LINKAGE IN R: TRY IT YOURSELF